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Extracting and Developing Nano Carboxymethyl Cellulose for Enhanced Drilling Fluid Formulation Using Local Resources

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Abstract

Cellulose-based additives serve as vital raw materials in oil and gas industry to resolve various issues encountered during drilling operations, including excessive fluid invasion and poor hole cleaning while offering environmentally friendly and sustainable solutions. The objective of this study is to extract and develop nano carboxymethyl cellulose (CMC) from local resources to enhance drilling fluid formulations. The specific goals included optimizing the extraction process, evaluating the performance of nano CMC in drilling fluids, and assessing the sustainability and cost-effectiveness of utilizing local resources. The process involved selecting of cellulose-rich source material such as date palm waste, subjected to pre-treatment, delignification, and bleaching to remove impurities. The extracted cellulose is further processed to obtain crystalline nanocellulose (CNC) through acid hydrolysis. Subsequently, carboxymethylation is performed by reacting the CNC with monochloroacetic acid, yielding nano CMC with enhanced solubility and functionality.

Throughout the process, various characterizations are conducted to assess the chemical composition, morphology, structural properties, and thermal stability of the extracted cellulose and synthesized nano CMC. These characterizations include X-ray diffraction, Fourier-transform infrared spectroscopy (FTIR), scanning electron microscope (SEM), thermogravimetric analysis (TGA), and degree of substitution. The study successfully yielded nano CMC from local resources, with FTIR confirming the presence of carboxymethyl groups, and XRD pattern showing more intense peaks in nano CMC, indicating a higher degree of crystallinity compared to the untreated cellulose. TGA results exhibited a high thermal degradation temperature of N-CMC compared to extracted cellulose, and SEM images revealed well-defined nanoscale structures. The optimization of the carboxymethylation process led to a degree of substitution (DS) of 0.9 to further enhance the functionality of the nano CMC.

Introduction

Cellulose-based materials are considered one of the most sustainable and environmentally friendly additives that has been used in various industries. Cellulose derivatives can provide high performance with lower cost, making them ideal candidate over synthetic polymers (Ramasamy, Amanullah 2019). In the realm of oil and gas, drilling fluids play a vital role in drilling operations, performing many functions such as providing hydrostatic pressure, cooling and lubricating the drill bit, suspending and transporting drilled cuttings to the surface, reducing the fluid flow into the porous zones to minimize formation damage. Carboxymethyl cellulose (CMC) is an anionic, water-soluble polymer derived from plant-based cellulose. CMC consists of a linear polysaccharide of endoglucase repeating units connected by β -1,4-glycosidic bonds (Rahman, 2021). CMC has the same backbone as cellulose with substituted group (-CH₂-COOH) on some of the hydroxyl (-OH) groups on the glucose units in the cellulose chain occurring primarily at the C2, C3, and/or C6 positions of the glucose units. (Rahman, 2021) it is renowned for its water solubility due to the presence of hydrophilic carboxymethyl group and ability to enhance the rheological characteristics of drilling fluids by increasing the viscosity which is crucial to the downhole cleaning, suspending of the drilled cuttings from the wellbore to the surface, and prevent any accumulation in the wellbore (Hughes et al., 1993). Several operational issues might occur in terms of poor rheological characteristics, including insufficient carrying capacity, inadequate solid suspension, poor hole cleaning. The consequences for such implications are increasing the risk of stuck pipe, increased operational cost, and reduced the efficiency of the drilling.

Traditionally, the majority of CMC used in drilling fluid formulations has been imported from overseas markets which leads to supply chain challenges and increased costs. However, with an increasing demand for well exploration more resources are required to support growing operations. The vast availability of cellulose-rich biomass, such as the date palm waste (phoenix dactyliferous) represents a promising abundant feedstock to develop locally sourced CMC. This approach not only aligns with the growing demand for sustainable materials but also addresses economic and environmental concerns by reducing dependency on imported products. Date palm trees are one of the most abundant biomass with 23 million trees in Saudi Arabia. Each year, more than 200,000 tons of biomass waste are generated from the trimming of palm branches. This huge biomass waste holds a potential for recycling and converting them to valuable alternatives to replace non-degradable products (Faiad et al., 2022). Plant-based materials also referred to as lignocellulosic biomass are generally composed of hemicellulose, lignin, and cellulose (Rahman et al., 2017). In order to process lignocellulosic and isolate the cellulose fibers, lignin and hemicellulose related components must be removed (Rahman et al., 2017). Once purified, the cellulose can be utilized in a wide range of products, including cellulose derivatives, such as carboxymethyl cellulose, hydroxyethyl cellulose, nanocellulose, and other bio-based materials to be used for different applications. The date palm is one promising local biomass resource that is rich in cellulose - the key raw material for CMC production.

The study aims to localized CMC by pretreatment of collected date palm trunk, including delignification using alkaline agent to remove lignin and hemicellulose, bleaching is carried out using sodium chlorite to obtain a purified cellulose-rich fraction, and acid hydrolysis to obtain cellulose nanocrystals with high crystalline structure (Raza et al., 2022) (Alhamzani & Habib, 2021). Subsequently, the extracted cellulose is then subjected to alkalization, typically with sodium hydroxide, followed by carboxymethylation using monochloroacetic acid to introduce carboxymethyl groups to develop CMC with properties tailored for drilling fluid applications (Khan et al., 2024). Degree of substitution is defined as the average number of hydroxyl groups that have been replaced by carboxymethyl groups per anhydroglucose unit (Casaburi et al., 2018). Degree of substitution is a unique property of CMC, influencing its viscosity, solubility and functionality (Rahman et al., 2021). This key parameter can be controlled by adjusting the reaction time, temperature, and reagent concentrations (Rahman et al., 2021) (Wahyuni et al., 2019). After the etherification step, the CMC product undergoes purification through washing and filtration to remove

unreacted chemicals and byproducts. Finally, the CMC is dried to the desired moisture content and characterized to ensure it meets the specifications required for use as a drilling fluid additive.

Materials and Methods

Materials

Date palm trunk waste was collected from local resources and used as the raw material for cellulose extraction. Sodium hydroxide (98%), sodium chlorite (NaClO_4), sulfuric acid (H_2SO_4), ethyl alcohol, acetic acid glacial, chloroacetic acid, and dialysis tube. All reagents were used without further purification and employed at varying concentrations throughout the reaction steps.

Material Preparation

The date palm trunk pieces were first washed thoroughly with distilled water to remove any impurities, dirt, or debris. The clean trunk was then dried in a sunny area for 24 hours to remove any residual moisture. After drying, grind with a mechanical grinder to a mesh size of 160 μm .

Extraction of Cellulose

Delignification. The trunk pieces were immersed in a 5% (w/v) sodium hydroxide (NaOH) solution and heated to 80°C under reflux. The mixture was stirred continuously for 2 hours using a magnetic stirrer. This process helped to solubilize and remove the non-cellulosic components, such as hemicellulose and lignin (Rahman et al., 2017). The mixture was filtered using a Büchner funnel, and the extracted cellulose was thoroughly washed with distilled water until the pH of the filtrate reached a neutral value (pH 7.0). The purified cellulose was then dried in a convection oven at 60°C for 24 hours to obtain the final cellulose product.

Bleaching. The extracted cellulose was then subjected to a bleaching process to remove any residual lignin and discoloration. The cellulose was treated with a 5% (w/v) sodium chlorite (NaClO_2) solution, and the pH was adjusted to 4-5 using 2% (v/v) acetic acid (CH_3COOH). The mixture was heated to 80°C and stirred for 2 hours. The bleached cellulose was then washed thoroughly with distilled water until the pH of the filtrate reached a neutral value. The purified cellulose was dried in a convection oven at 60°C for 24 hours.

Extraction of Cellulose Nanocrystals

The bleached cellulose was further processed to obtain cellulose nanocrystals (CNCs) and adjusted from previous studies (Alhamzani, & Habib, 2021) (Raza et al., 2022). The cellulose was dispersed in a 30% (v/v) sulfuric acid (H_2SO_4) solution and the suspension was heated to 45 °C. The mixture was stirred continuously for 45 minutes to facilitate the acid hydrolysis of the cellulose, which selectively removes the amorphous regions and leaves behind the nano-sized crystalline domains (Ramasamy & Amanullah, 2019). The reaction was quenched by diluting the mixture with cold distilled water. The CNCs were separated from the reaction mixture by centrifugation at 8000 rpm for 10 minutes. The obtained CNCs were then washed repeatedly with distilled water until a neutral pH was achieved. The purified CNC suspension was then subjected to dialysis for 5 days, with the water being changed daily, to remove any residual sulfuric acid. The purified CNC suspension was subsequently dried in a freeze-dryer to obtain the final CNC powder.

Development of Nano Carboxymethyl Cellulose

Alkalization. The dried cellulose nanocrystals (CNCs) were activated by dispersing them in a 35% (w/v) sodium hydroxide (NaOH) solution. The mixture was stirred at 30°C for 1 hour to convert the hydroxyl groups on the CNC surface into alkoxide groups (Wahyuni et al., 2019).

Carboxymethylation. The method for developing CMC was adopted from previous studies (Ab Rasid et al., 2021). The extracted cellulose was then mixed in monochloroacetic acid (ClCH₂COOH) with molar ratio of cellulose to MCA is 1:2. The reaction mixture was heated to 60°C and stirred for 3 hours to facilitate the carboxymethylation of the extracted cellulose. After the carboxymethylation reaction, the mixture was cooled to room temperature, and the modified nano carboxymethyl cellulose (nano-CMC) was separated by centrifugation at 10,000 rpm for 10 minutes. The nano-CMC was then washed repeatedly with 70 ethanol to remove any unreacted reagents and impurities, including sodium chloride, sodium glycolate, and by-products (Altunina et al., 2016). Finally, the purified nano-CMC was freeze-dried to obtain the final product as a powder.

Characterization

Fourier Transform Infrared Spectroscopy (FTIR)

The Fourier Transform Infrared Spectroscopy (FTIR) analysis was conducted to examine the chemical structural changes in raw cellulose, extracted cellulose and nano CMC. Nicolet iS10 instrument in transmission mode was used to record the FTIR spectrums. The selected parameters were the average scans of 16 per sample, a resolution of 4 cm⁻¹, and the transmission range for the analysis was 400 - 4000 cm⁻¹. The FTIR spectrometer was purged with nitrogen gas to eliminate interference of atmospheric gases. A prepared potassium bromide (KBr) pellet was used as a background and subtracted from the sample's spectrum.

Thermogravimetric Analysis (TGA)

The thermal stability profile and decomposition behavior of the samples was evaluated using thermogravimetric Analysis (TGA) using a TA Discovery 5500 instrument. About 5-10 mg of each sample was placed in a platinum pan and heated for maximum temperature up to 800 °C. The samples were heated at a constant rate of 10 °C/min under continuous flow of nitrogen atmosphere at flow rate of 40 mL/min.

X-Ray Diffraction (XRD)

X-ray diffraction (XRD) spectroscopy was carried out to confirm the crystalline structure using Shimadzu XRD-7000 X-ray diffractometer. Cu-K radiation was used as source to examine the samples with a voltage of 40 kV, an electrical current of 30 mA. The diffraction data were collected from 10° to 80° 2θ Bragg angles with a step size of 0.02° at a scan rate of 2°/minute. The samples were ground with motor and pestle to ensure homogenies surface and packed into standard sample holder. The crystallinity index of the samples was calculated using the Segal equation, as illustrated in equation (1).

$$CrI(\%) = \frac{I_{002} - I_{am}}{I_{002}} \times 100 \quad (1)$$

where I₀₀₂ denotes the intensity of the crystallinity peak (usually near 2θ = 22°) and I_{am} denotes the intensity of the amorphous region (usually near 2θ = 16°).

Scanning Electron Microscopy (SEM)

The surface morphology of the synthesized cellulose nanocrystals was examined using a field emission scanning electron microscope. Imaging was performed at an accelerating voltage of 3 kV, with a probe current of 30 pA and a working distance of 17 mm. To prevent electrostatic charging during the imaging process, all samples were coated with a thin 3 nm layer of iridium.

Degree of Substitution (DS)

Weigh about 0.5 g of dried CMC in a porcelain dish to be placed in a muffle furnace, and slowly increase the temperature in increments of 100 °C. Once the temperature reaches 400 °C, open the oven door carefully to confirm that the material has charred. Increase the temperature to 650 °C, maintain it for 45-50 minutes,

and then stop the heating. Let the furnace cool to at least 400 °C, and then slightly open the oven door, taking extreme caution due to the heat hazard, as it is still at a very high temperature. When the temperature is about 200 °C, remove the dish using a long pair of tongs and while wearing heat-resistant gloves. After cooling, add about 20 mL of distilled water while stirring on a magnetic stirrer to allow all the residue to dissolve completely. Titrate with 0.1 N sulfuric acid using methyl red as an indicator. If no sharp endpoint is observed, the solution must be boiled to expel the dissolved carbon dioxide, and continue titration to ensure the endpoint is sharp and stable. The degree of substitution is conducted using [equation \(2\)](#).

$$\text{Degree of Substitution} = \frac{0.162x B}{1 - 0.08B} \quad (2)$$

Where $B = \frac{0.1xb}{c}$, $B =$, B = volume of standard sulfuric acid, & c = weight of purified CMC taken for ignition

Electromagnetically Spinning (EMS) Viscometer

The viscosity measurements of nano CMC were conducted using an Electromagnetically Spinning (EMS) viscometer. The EMS viscometer operates by inducing the rotation of a small magnetic sphere inside the sample, with the rotation rate controlled by an applied electromagnetic field. For this study, the viscosity of the samples was measured at various rotational speeds, specifically at 1000, 600, 300, 200, and 100 revolutions per minute (RPM), at a controlled temperature of 25°C. A 2% CMC solution was prepared by gradually adding 2 g of CMC to 100 ml of DI water. The mixture was thoroughly stirred until the CMC was completely dissolved, forming a gel-like solution as depicted in [Figure 5](#).

API Filter Press Fluid Loss & FANN 35 Viscometer

The viscosity of the drilling fluid was measured using a FANN 35 viscometer to determine its rheological properties at varying shear rates (600, 300, 200, 100, 6, and 3 rpm). The API filter press fluid loss test was also conducted to evaluate the filtration control of the mud. For water-based mud preparation, 350 mL of water was mixed with 1 g of xanthan gum, 0.15 g of sodium hydroxide, and 6 g of N-CMC. The N-CMC was added slowly at a uniform rate while stirring on the mixer to ensure thorough dispersion for optimal viscosity. A blank base formulation was also prepared without CMC additives. The second formulation included N-CMC, while the third formulation incorporated the equivalent commercial CMC product to allow for performance comparison. The formulation used is as follows in the [table 1](#) below.

Table 1—Drilling Fluid Formulation Used for Testing CMC Performance

Additive, g	Blank mud	Mud + N-CMC	Mud + commercial product
Water	350	350	350
XC-Polymer	1	1	1
N-CMC	0	6	0
Equivalent commercial product	0	0	6
NaOH	0.15	0.15	0.15

Results and Discussion

Fourier Transform Infrared Spectroscopy (FTIR)

To confirm the successful removal of lignin and hemicellulose related components, FTIR is conducted as depicted in [Figure 1](#). The raw date palm waste showed various absorption bands of the lignocellulosic structure composition such as cellulose, lignin and hemicellulose. Lignin as well as hemicellulose structure bands were apparently located within the region of 1700-1750 cm⁻¹ corresponding to the C=O stretching

of acetyl groups in hemicellulose. Apart from this, the band in the range of 1230-1260 cm^{-1} was attributed to the ether functional group C-O-C, which was related to lignin (Rahman et al., 2017). These lignin and hemicellulose associated bands were significantly reduced after bleaching and delignification treatments, indicating the elimination of lignin and hemicellulose. For Na-CMC, the characteristic peak is the C=O stretching at 1600-1650 cm^{-1} indicating the presence of carboxymethyl groups that have been introduced onto cellulose (Li et al., 2023). All samples exhibit a very large broad band O-H stretching at the range of 3200-3600 cm^{-1} and a C-H stretching band at around 2800-3000 cm^{-1} suggesting the cellulose based materials being hydrophilic in nature.

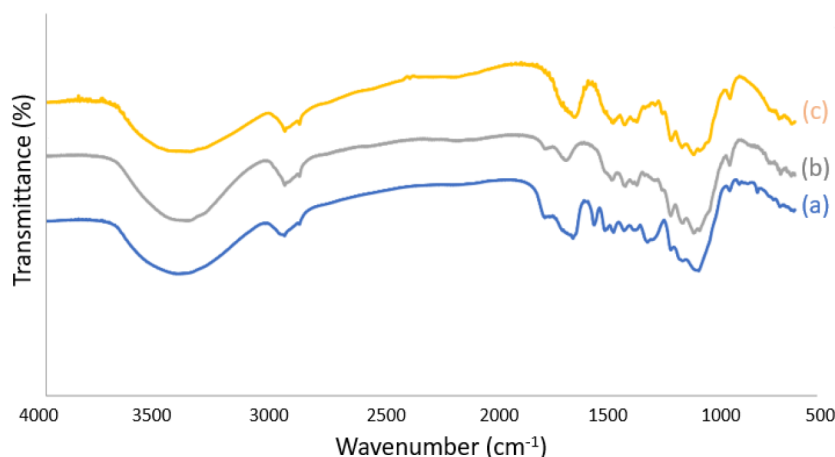


Figure 1—Fourier transform infrared spectrometry of (a) R-Cellulose, (b) Extracted Cellulose, and (c) CMC Nanocrystal.

Thermogravimetric Analysis (TGA)

The thermal profiles obtained from TGA data for the three samples are illustrated in Figure 2. The results revealed that the examined samples have different thermal behaviors. All samples began to lose weight from the beginning of the run, the weight loss in the region between 25 to 150 °C can be assigned to the moisture evaporation. Significant thermal degradation occurs between 250 °C to 400 °C of row cellulose, corresponding to the breakdown of the cellulose polymer chains, resulting in a weight loss of approximately 79.59% at 800 °C and a remaining residue of 20.40%, primarily inorganic ash. Extracted cellulose shows main thermal decomposition similar to raw cellulose, around between 250°C to 400°C, but with a lower weight loss compared to row cellulose. The total weight loss measured is 76.79% with finale ash content of 23.20%, indicating a purification effect from the alkalization and bleaching process (Raza et al., 2022). The major thermal degradation of nano CMC occurs within a slightly lower temperature range of 250 °C to 350 °C, with a total weight loss of 74.70% and 25.28% residual weight after complete combustion. This indicates that the removal of the amorphous structure and the presences of carboxymethyl groups enhance the degradation process compared to untreated cellulose.

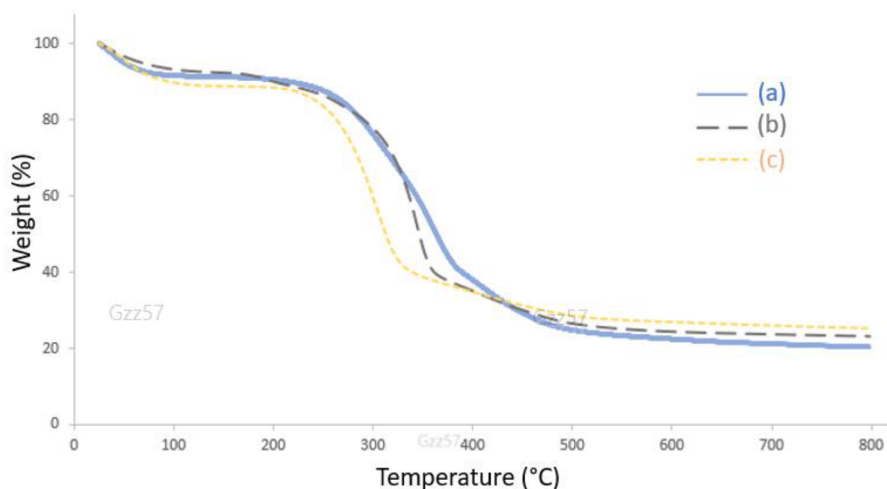


Figure 2—TGA curve of (a) R-Cellulose, (b) Extracted Cellulose, and (c) CMC Nanocrystal.

X-Ray Diffraction

X-ray diffraction (XRD) analysis is performed to assesses the crystalline structure of raw cellulose, extracted cellulose, and CMC nanocrystals. All respective samples displayed three peaks at approximately $2\theta=16^\circ, 22^\circ$, & 35° , corresponding to (110), (200), and (004). XRD pattern of raw cellulose typically contain two broad peaks located at $2\theta = 16.68^\circ$ and 22.08° and 31.84° , indicating the presence of both an amorphous and a crystalline structure, with a minimum crystalline index of 55%. Moreover, the cellulose obtained after delignification demonstrates peaks at about 22.46° and 32.15° , indicating the reduction of the amorphous region after delignification and bleaching treatments giving a better crystallinity of 60%. Also, CMC has crystalline peaks at 27.15° degrees and again at 35.24° indicating that the crystallinity of cellulose nanocrystals has been enhanced because of the successful removal of hemicellulose and lignin after acid hydrolysis.

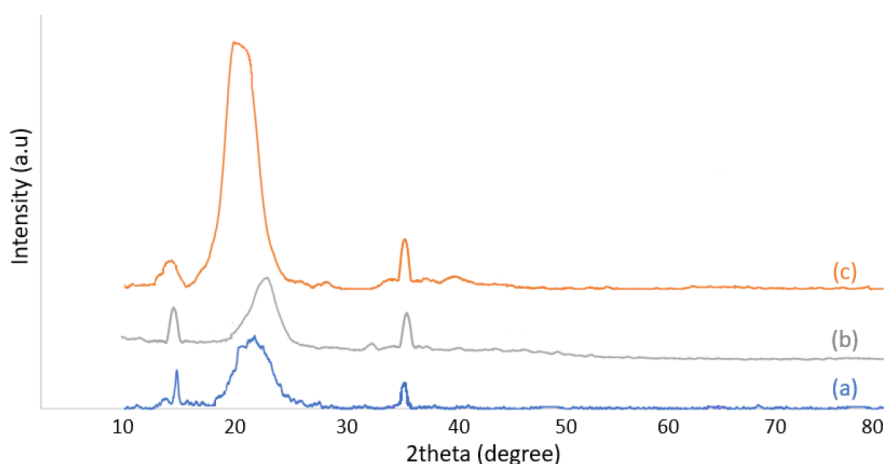


Figure 3—X-ray diffractions of (a) R-Cellulose, (b) Extracted Cellulose, and (c) CMC Nanocrystal.

Morphological Analysis

The SEM image reveals a detailed topographical map of the CMC nanocrystals, showcasing their unique morphology. Various sizes are observed, with some nanocrystals appearing needle-like or elongated, and others showing more irregular shapes. Clusters of agglomerated particles are also visible, indicating interactions among the nanocrystals. These observations, as depicted in Figure 4, provide a view of the surface morphological structure and characteristics of the nano CMC.

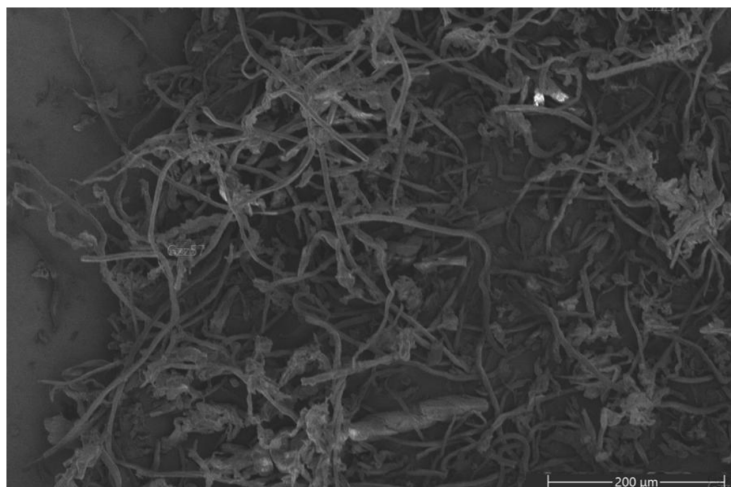


Figure 4—Scanning electron microscopic images of CMC Nanocrystal.

Degree of Substitution

The degree of substitution (DS) for the synthesized nano CMC was determined to be 0.9, indicating that on average 0.9 hydroxyl groups per anhydroglucose unit (AGU) in the cellulose structure were successfully replaced by carboxymethyl groups. This DS value is favorable for drilling fluids application, as it ensures effective thickening and stabilization properties without excessively increasing viscosity that might lead to negatively impact the rate of penetration.

Viscosity

The viscosity results for the 2% nano CMC solution at 25°C are summarized in Table 2. The measurements indicate a shear-thinning behavior (Rahman et al., 2021), as the viscosity decreases with increasing rotational speed. The 2% CMC solution prepared resulted in a gel-like consistency, as shown in Figure 5.

Table 2—Thermogravimetric analysis of (a) R-Cellulose, (b) Extracted Cellulose, and (c) CMC Nanocrystal.

Sample	Total weight loss at 800°C, %	Residual masses at 800°C, %
Raw cellulose	79.59	20.40
Extracted cellulose	76.79	23.20
N- CMC	74.70	25.28

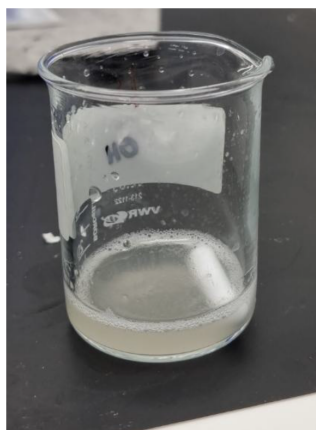


Figure 5—2% CMC solution showing gel-like consistency.

Performance Testing

The drilling fluids prepared exhibited performance variations based on the additive used. The base mud showed limited rheological functionality, with a plastic viscosity (PV) of 4 cP and yield point (YP) of 9 lb/100 ft², which suggests low to moderate suspension capability but insufficient fluid loss control. Gel strengths at 10 seconds and 10 minutes (3 and 4 lb/100 ft², respectively) were low, indicating weak structural development during static periods. Additionally, the formulation failed to control fluid loss under API conditions. The N-CMC-based formulation presented a noticeable improvement in both rheological and filtration properties. With a PV of 8 cP and YP of 11 lb/100 ft², the mud maintained a balanced viscosity profile, supporting effective cuttings transport. Gel strength values (4 and 5 lb/100 ft²) were slightly higher than the base mud. Importantly, the fluid loss was reduced to 11.2 ml suggesting that the combination of N-CMC and xanthan gum contributed to forming a relatively low-permeability filter cake. The formulation containing the commercial equivalent product showed higher viscosity parameters (PV: 12 cP, YP: 15 lb/100 ft²) and stronger gel strengths (7 and 9 lb/100 ft²). The filtrate volume was lower at 8 ml, suggesting more effective filtration control in this case. The overall performance of each formulation is summarized in [Table 4](#).

Table 3—Viscosity measurements of CMC Nanocrystal.

Viscometer dial reading, rpm	N-CMC (2% Solution), cP
1000	72.14
600	76.04
300	81.88
200	81.84
100	101.08

Table 4—Performance testing results of CMC Nanocrystal.

Tests	Parameters	Blank mud	Mud + N-CMC	Mud + commercial product
Fluid rheology	Dial reading at 600 rpm, cP	17	27	39
	Dial reading at 300 rpm, cP	13	19	27
	Dial reading at 200 rpm, cP	11	16	21
	Dial reading at 100 rpm, cP	8	12	19
	Dial reading at 6 rpm, cP	3	4	8
	Dial reading at 3 rpm, cP	2	3	6
	Plastic Viscosity, cP	4	8	12
	Yield Point, lb/100 ft ²	9	11	15
	Gel Strength 10min/10 sec, lb/100 ft ²	3/4	4/5	7/9
Fluid loss	Filtrate Volume, ml	No control	11.2	8

Conclusion

The isolation and development of nanocellulose carboxymethyl cellulose (nano CMC) from local date palm waste have been successfully achieved. The developed nano CMC exhibits improved performance characteristics, making it a valuable additive for drilling fluids in the oil and gas industry. Its enhanced properties include better thickening and stabilization effects, which are crucial for optimizing drilling

operations. The prepared drilling fluid provided balanced rheological properties and controlled fluid loss, enhancing the drilling process and ensuring operational stability in the wellbore. Furthermore, the use of date palm waste as a source material highlights the sustainability of this approach. By utilizing local biomass, the process not only reduces waste but also provides an environmentally friendly alternative to conventional petroleum-based additives. This development aligns with broader industry trends towards greener and more sustainable practices.

References

- Ab Rasid, Nurul & Mohammad Zainol, Muzakkir & Saidina Amin, Nor Aishah. (2021). Synthesis and Characterization of Carboxymethyl Cellulose Derived from Empty Fruit Bunch. *Sains Malaysiana*. **50**. 2523–2535. [10.17576/jsm-2021-5009-03](https://doi.org/10.17576/jsm-2021-5009-03).
- Alhamzani, A.G., & Habib, M.A. (2021). Preparation of cellulose nanocrystals from date palm tree leaflets (Phoenix dactylifera L.) via repeated chemical treatments. [https://www.semanticscholar.org/paper/Preparation-of-cellulose-nanocrystals-from-date-palm-tree-leaflets-\(phoenix-dactylifera-l.\)-via-repeated-chemical-treatments.-Alhamzani-Habib/046cee7b1670f1a666c611c759faed955d3e2d9c](https://www.semanticscholar.org/paper/Preparation-of-cellulose-nanocrystals-from-date-palm-tree-leaflets-(phoenix-dactylifera-l.)-via-repeated-chemical-treatments.-Alhamzani-Habib/046cee7b1670f1a666c611c759faed955d3e2d9c)
- Altunina, L. & Tikhonova, L.D. & Yarmukhametova, E.G. (2016). Method for Deriving Carboxymethyl Cellulose. *Eurasian Chemico-Technological Journal*. **3**. 49. [10.18321/ectj416](https://doi.org/10.18321/ectj416).
- Casaburi, A., Rojo, Ú.M., Cerrutti, P., Vázquez, A., & Foresti, M.L. (2018). Carboxymethyl cellulose with tailored degree of substitution obtained from bacterial cellulose. *Food Hydrocolloids*, **75**, 147–156. <https://doi.org/10.1016/j.foodhyd.2017.09.002>
- Faiad, A.; Alsmari, M.; Ahmed, M.M.Z.; Bouazizi, M.L.; Alzahrani, B.; Alrobei, H. Date Palm Tree Waste Recycling: Treatment and Processing for Potential Engineering Applications. *Sustainability* 2022, **14**, 1134. <https://doi.org/10.3390/su14031134>
- Hughes, T.L., Jones, T.G.J., & Houwen, O.H. (1993). Chemical characterization of CMC and its relationship to drilling-mud rheology and fluid loss. <http://pascal-francis.inist.fr/vibad/index.php?action=getRecordDetail&idt=4895973>
- Khan, M.A., Li, M., Lv, K., Sun, J., Liu, C., Liu, X., Shen, H., Dai, L., & Lalji, S.M. (2024). Cellulose derivatives as environmentally-friendly additives in water-based drilling fluids: A review. *Carbohydrate Polymers*, **342**, 122355. <https://doi.org/10.1016/j.carbpol.2024.122355>
- Li, Ang & Zheng, Xudong & Ji, Biao & Xu, Tongtong & Zhang, Yuzhe & Rong, Jian & Mei, Jinfeng & Zhou, Man & Li, Zhong-Yu. (2023). Modified carboxymethyl cellulose nanocrystalline imprinted aerogels with three-dimensional interoperable pore structure for selective adsorption of gadolinium. [10.21203/rs.3.rs-2558432/v1](https://doi.org/10.21203/rs.3.rs-2558432/v1).
- Ramasamy, J., & Amanullah, M. (2020). Nanocellulose for oil and gas field drilling and cementing applications. *Journal of Petroleum Science and Engineering*, **184**, 106292. <https://doi.org/10.1016/j.petrol.2019.106292>
- Raza, M., Abu-Jdayil, B., Banat, F., & Al-Marzouqi, A.H. (2022). Isolation and Characterization of Cellulose Nanocrystals from Date Palm Waste. *ACS Omega*, **7**(29), 25366–25379. <https://doi.org/10.1021/acsomega.2c023>
- Rahman, M.S., Hasan, M.S., Nitai, A.S., Nam, S., Karmakar, A.K., Ahsan, M.S., Shiddiky, M.J.A., & Ahmed, M.B. (2021). Recent developments of carboxymethyl cellulose. *Polymers*, **13**(8), 1345. <https://doi.org/10.3390/polym13081345>
- Rahman, N.H.A., Chieng, B.W., Ibrahim, N.A., & Rahman, N.A. (2017). Extraction and Characterization of Cellulose Nanocrystals from Tea Leaf Waste Fibers. *Polymers*, **9**(11), 588. <https://doi.org/10.3390/polym9110588>
- Faiad, A.; Alsmari, M.; Ahmed, M.M.Z.; Bouazizi, M.L.; Alzahrani, B.; Alrobei, H. Date Palm Tree Waste Recycling: Treatment and Processing for Potential Engineering Applications. *Sustainability* 2022, **14**, 1134. <https://doi.org/10.3390/su14031134>
- Wahyuni, Henny & Yuliasmi, Sri & Aisyah, Hanifah & Riati, Devi. (2019). Characterization of Synthesized Sodium Carboxymethyl Cellulose with Variation of Solvent Mixture and Alkali Concentration. *Open Access Macedonian Journal of Medical Sciences*. **7**. 3878–3881. [10.3889/oamjms.2019.524](https://doi.org/10.3889/oamjms.2019.524).